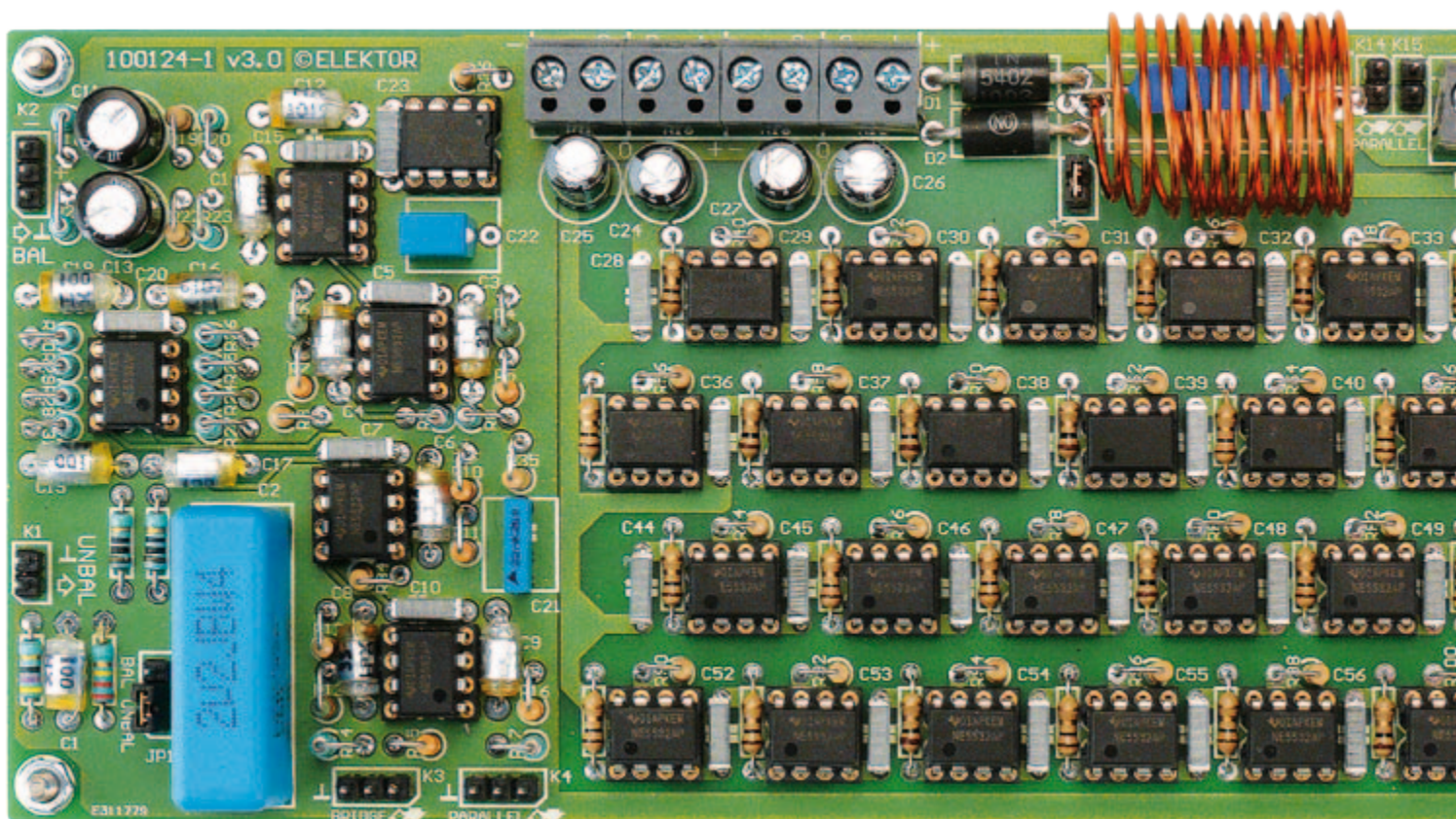


The 5532 OpAmp

Part 1: design philosophy and schematics



By Douglas Self (UK)

The most popular dual opamp in the world of audio is the (NE)5532. An interesting power amplifier can be made by connecting enough 5532s in parallel, how about 32 for a start? This may sound like a radical course of action, but it actually works very well, making it possible to build a very simple amplifier that retains not only the excellent linearity but also the power-supply rejection and the inbuilt overload protection of the 5532, which reduces the external circuitry required to a minimum.

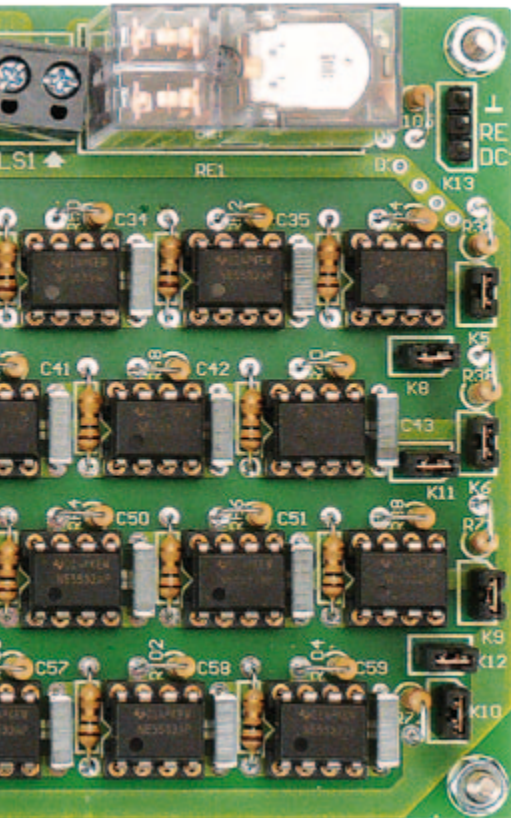
While not exactly a brand-new design, the type (NE)5532 dual operational amplifier (opamp) is a very capable device giving low distortion with good load-driving capabilities, and a remarkably good noise performance. It is only quite recently that better opamps for audio work have become available. While these can give truly outstanding results, the cheapest of them costs ten times more than the 5532, which is available at a remarkably

low price — in fact it is one of the cheapest opamps, because it is so widely used in audio applications.

It should be mentioned at once that the obvious limitation with using opamps to drive loudspeakers is that the output voltage swing is limited compared with a conventional power amplifier, and using a single-ended array of 5532s will give about 15 W_{rms} into 8 Ω. This output can be greatly extended by using two such amplifiers in

bridge mode; one amplifier is driven with an inverted input signal so the voltage difference between the two amplifier outputs will be doubled, and the power output is quadrupled to about 60 W_{rms} into 8 Ω. This should be enough for most domestic hifi situations.

The other unalterable limit set by the opamps is the maximum output current, set by the internal overload protection. A single 5532 section (one half of the dual package)



Specification — per channel, 8 ohm load

Supply voltage	±18.3 V	
Input sensitivity	- unbalanced	840 mV (16 W, 1 % THD)
	- balanced	833 mV (16 W, 1 % THD)
Input impedance	- unbalanced	38.8 kΩ
	- balanced	93.6 kΩ
Output power, sinewave	- 0.1 % THD	16 W
	- 1 % THD	16.8 W
Output power bandwidth		1.5 Hz – 275 kHz
Slew rate		5 V/μs
Rise time		4 μs
Signal/noise ratio	(1 W ref.)	110 dBA
	- 108 dB (B = 22 Hz – 22 kHz linear/unweighted)	
Harmonic distortion + noise	- 0.0005% (B = 22 kHz, 1 kHz, 1 W)	
	- 0.0009% (B = 80 kHz, 1 kHz, 1 W)	
	- 0.0004% (B = 22 kHz, 1 kHz, 8 W)	
	- 0.0005% (B = 80 kHz, 1 kHz, 8 W)	
	- 0.003 % (B = 80 kHz, 20 kHz, 8 W)	
Intermodulation distortion	- 0.0012% (1 W)	
	- (50 Hz : 7 kHz = 4 : 1) 0.0015% (8 W)	
Dynamic IM distortion	- 0.0011% (1 W)	
	- (3.15 kHz square wave + 15 kHz sine wave)	0.0035% (8 W)
Damping factor	- 194 (1 kHz)	
	- 111 (20 kHz)	
DC-protection	±1.5 V	
Quiescent current	300 mA	

will drive 500 Ω to the full voltage output, though it is advisable to keep the loading lighter than this to maintain low distortion at high levels. If 4 Ω operation is required, twice as many opamps must be used to supply the doubled current demand. This also applies to bridged operation into 8 Ω. The system is designed so that either single-ended or bridged operation can be used; the basic design described here gives a working stereo amplifier with just three PCBs. The amplifier cards can be paralleled without problems, and facilities are provided to connect more PCBs in parallel for driving low-impedance speakers.

Overload protection is inherent in the opamps, but output relays are used for on/off muting and to protect loudspeakers against a DC fault.

A tour of the design

The schematic in **Figure 1** shows one channel of the complete amplifier, which consists of

unbalanced and balanced line inputs, and the power amplifier itself, which is divided into a +22.7 dB gain stage and an array of paralleled output opamps configured as voltage-followers, giving the maximum amount of negative feedback around them to minimise distortion. Let's have a look at the various sections of the circuit.

The unbalanced input

This consists simply of RF filter R1, C1 and DC-drain R2, which are directly connected to the gain stage when JP1 is in the 'unbalanced' position.

The balanced input

This amplifier is an innovative design that gives very low noise. The conventional balanced input stage built with four 10 kΩ resistors and a 5532 opamp has a far worse noise performance than a simple unbalanced input, and is also much noisier than most power amplifiers; output noise is approximately –104 dBu. This balanced

amplifier here solves this problem partly by the use of a dual balanced stage (IC5A, IC5B) amplifier that partially cancels the uncorrelated noise from each amplifier, giving a 3 dB noise reduction, and in a similar way improves the CMRR; it also uses much lower resistor values than usual (820 Ω instead of 10 kΩ) which produces less Johnson noise in the first place. This is only possible because it is driven by unity-gain buffers IC4A, IC4B, which also allow the input impedances to be much higher than usual, preventing loading of external equipment and further improving the CMRR. The noise output is less than –112 dBu, an 8 dB improvement over conventional technology.

The gain stage

The main input amplifier is another innovative design that achieves very low distortion by spreading the gain required over three stages. +22.7 dB could easily be obtained with one opamp but 5532s are not

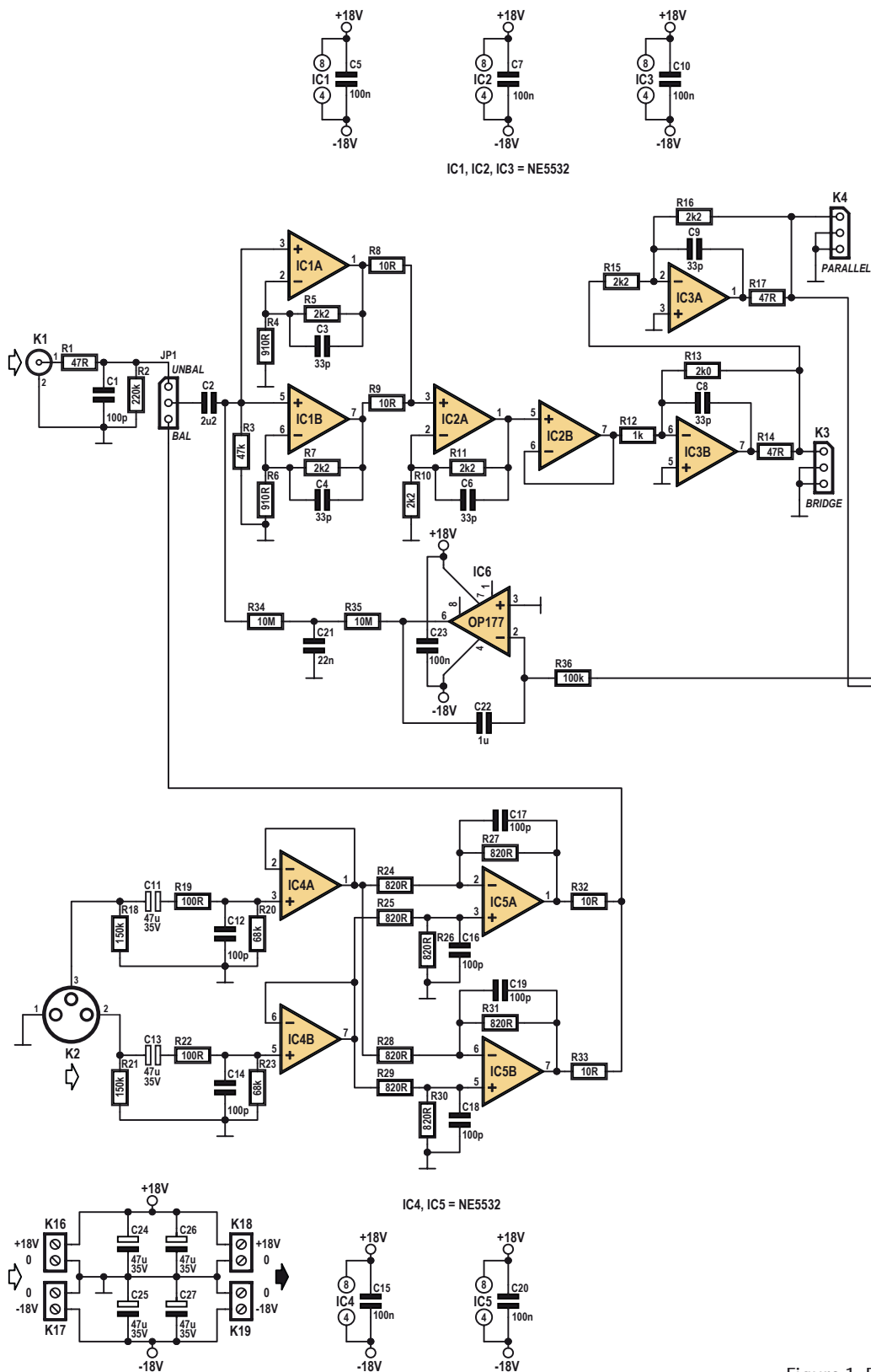
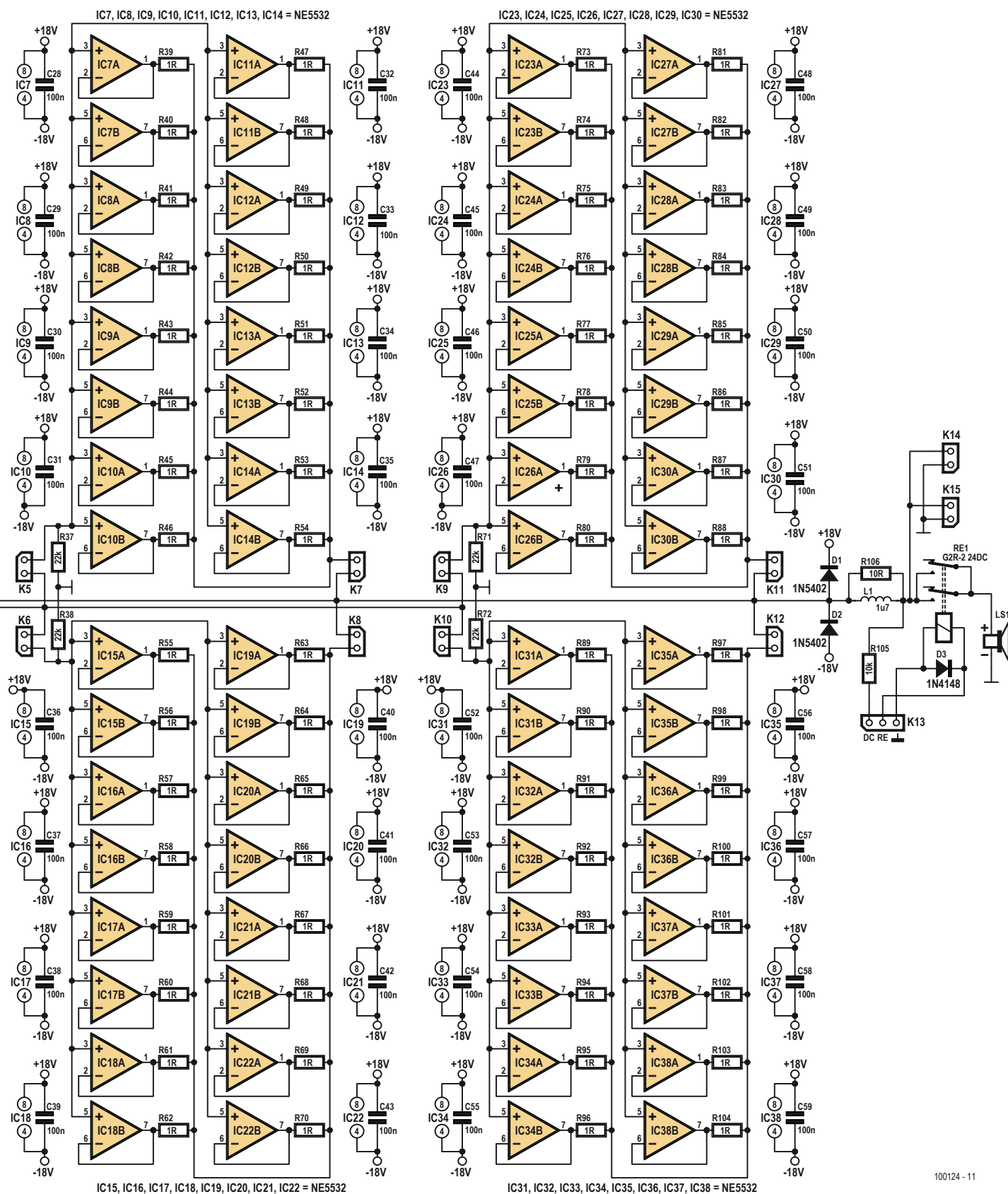
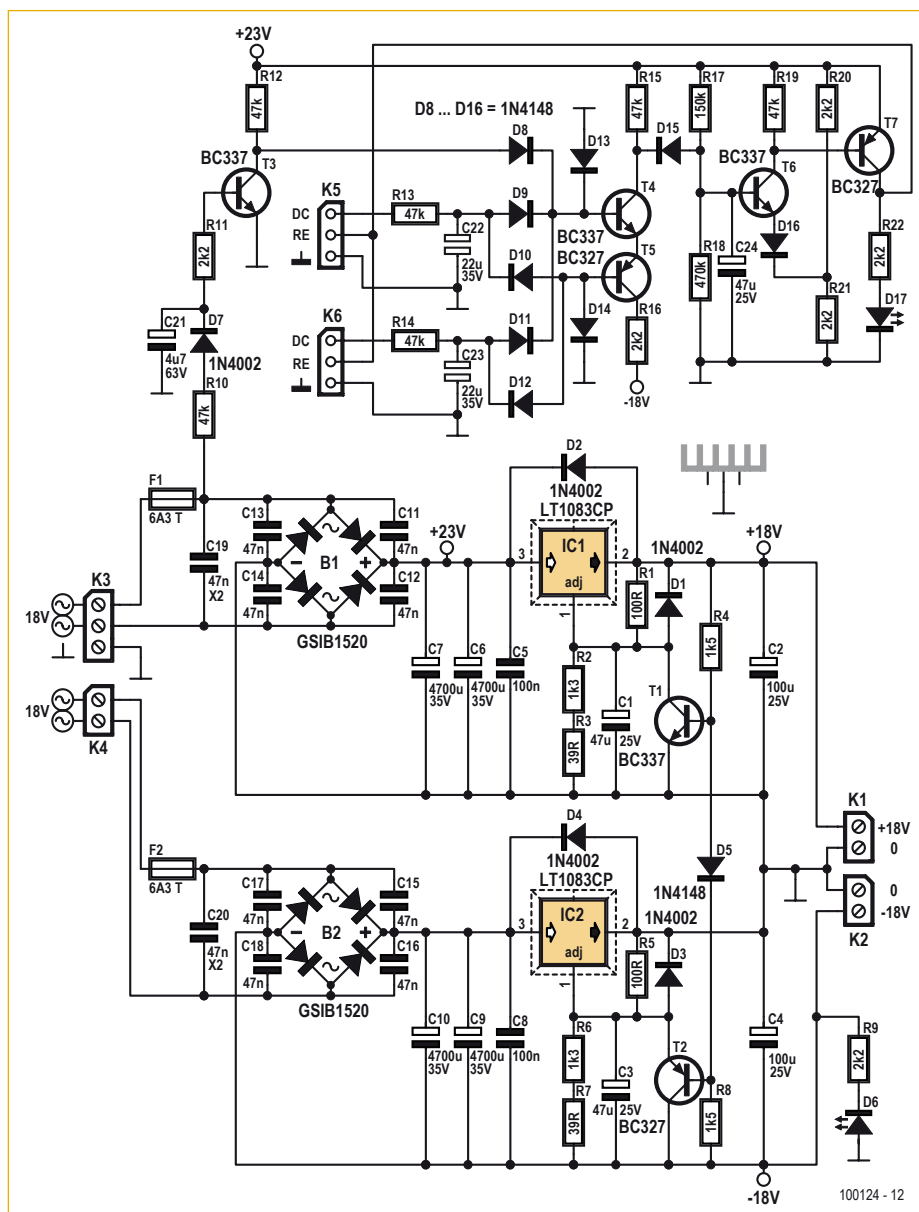


Figure 1. Power is indeed in numbers: circuit diagram of the basic NE5532 audio amplifier (one channel shown).





outputs joined by 1 Ω current-sharing resistors. These combining resistors are outside the 5532 negative-feedback loops, and you might wonder what effect they will have on the output impedance of the amplifier. A low output impedance is always a good thing, but not because of the so-called 'damping factor' which is largely meaningless as the speaker coil resistance always dominates the circuit resistance. 'Damping factor' is defined as load impedance divided by output impedance; we have 64 times 1 Ω resistors in parallel, giving an overall output impedance of 0.0156 Ω . This gives a theoretical damping factor of $8 / 0.0156 = 512$, very good by any standards. The wiring to the loudspeaker sockets will have more resistance than this!

The output opamps may be directly soldered into the board to save cost and give better conduction of heat from the opamp package to the copper tracks. However, on the prototype built in the Elektor labs, high quality sockets were used. Having a lot of opamps in parallel could make fault-finding difficult – if there is one bad opamp out of 32 then you are likely to have to do a lot of unsoldering (or IC unplugging) to find it. The opamp array is therefore split up into four sections of eight opamps, which are joined together by jumpers K5–K12, so on average you would only need to unsolder (or pull out) four opamps to find a defective one. In my many years of experience with it the 5532 has proven a very reliable opamp, and I think such failures will be very rare indeed. There is an output choke L1 for stability into capacitive loads, and catching diodes D1–D2 to prevent damage from voltage transients when current-limiting into reactive loads.

The output relay and its control

The output mute relay RE1 protects the loudspeakers against a DC offset fault and gives a slow-on, fast-off action so no transients are passed to the loudspeakers at power-up or power-down. The relay is controlled from the power supply board. With reference to **Figure 2**, at power-up R17 charges C24 slowly to give a turn-on delay. In operation C21 is charged and

completely distortion-free, and the THD would be significant.

The first stage (IC1A, IC1B) gives +10.7 dB of gain; the two outputs are combined by R8, R9 to give a 3 dB noise advantage, as in the balanced amplifier. The second stage IC2A gives +6 dB of gain. The gain is less to maximise negative feedback because the signal level is now higher. IC2B is a unity-gain buffer which prevents the 1 k Ω input impedance of final gain stage IC3B from loading the output of IC2A and causing distortion. IC2B is less vulnerable to loading because it has maximal negative-feedback. IC3B gives the final +6 dB of gain; it is used in shunt-feedback mode to avoid the

common-mode distortion which would otherwise result from the high signal levels here. It has a 'zero-impedance' output, with HF feedback via C8 but LF feedback via R13, so crosstalk is kept to a minimum while maintaining stability with load capacitance. The output at K3 is phase-inverted and can be used for bridging.

IC3A is a unity-gain inverting stage which corrects the signal phase. The output is also of the 'zero-impedance' type.

The power amplifier

The power amplifier consists of thirty-two 5532 dual opamps (i.e. 64 opamp sections) working as voltage-followers, with their

T3 is on; when the AC power is removed C21 discharges rapidly, T3 turns off, and D8 turns on T4–T5, which discharge C24 and cause the output relay contacts to be opened immediately. Even a brief AC power interruption gives the full turn-on delay. Normally T4 and T5 are off and D15 non-conducting, but if a DC offset fault applies either a positive or negative voltage via R13 or R14, T4–T5 turn on and the relays are opened at once to protect the loudspeakers.

Power supply

Again referring to the circuit diagram of the power supply unit in Figure 2, the $\pm 18\text{ V}$ symmetrical supply is regulated by two type LT1083 TO3-P positive regulators. When a 5532 sees one supply rail disappear, this opamp can get into an abnormal state in

which it draws excessive current. This could obviously be catastrophic with this design, so the PSU incorporates a mutual-shutdown facility which shuts off each supply rail if the other has collapsed due to short-circuiting or any other cause. If the positive rail collapses, T2 turns on and disables the negative supply. If the negative rail collapses, T1 turns on and disables the positive supply.

Cost

This project uses quite a lot of 5532s; 37 in each channel, but that does not mean the cost is excessive. In Great Britain, 5532s can be had from Rapid [1] for 24 p each at 100-off (Rapid are prepared to deal with anyone who has a credit card) This means that the cost of all the opamps would be about £18.00.

To be continued

Next month's closing instalment will cover approaches to constructing the amplifier on circuit boards, some performance figures obtained from our high-end test equipment, and an outline of challenges to those of you wishing to modify the amplifier for higher output powers and/or lower output impedance. Meanwhile this month's *E-Labs Inside* section has a page or so on issues with electrolytic capacitors encountered while the first prototype of the amplifier was tested.

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The 5532 OpAmp Amplifier (2)

Part 2: construction, bridged operation and test results

By Douglas Self (UK) and Ton Giesberts (Elektor Labs)

In this second and closing instalment we get real with two times 32 NE5532 opamps paralleled to form a high-end audio power amplifier with eminent specifications in terms of distortion and general sonic performance. There are also challenges for you: bridged operation for higher output power, and modifying the amplifier for 4-ohm operation.

If you thought that paralleling a few dozen NE5532 opamps is a curious way of designing a high-end audio amplifier and typical of Elektor's off-the-beaten track approach to electronics you are probably right. Last month's design considerations did not fail to trigger responses from you, our readership, in particular from all and sundry aspiring or even claiming to be a high-end audio designer. This month we get real by building the 5532 OpAmp Amplifier project and putting it through its paces.

Construction — amp board

The amplifier proper is built on a double-sided through plated circuit board of which the component overlay is shown in Figure 1. Two of these boards are required for a stereo amplifier. Boards supplied by Elektor come with through plating, a solder mask and holes predrilled. As such they are the best guarantee for success in replicating the project. While on the subject of quality, you are looking at an expensive, high-end amplifier. If you use mishmash components and *ditto* assembly methods and tools, you'll get mishmash results. More on selecting the best NE5532 brand to use in the inset.

Construction on the 205 × 84 mm board should not present problems as only standard through-hole parts are used. A properly functioning amp can be expected if you work with care and precision, like Ton Giesberts of Elektor Audio Labs who designed, built and tested all the boards pictured. Some notes and caveats deserve mention- ing, however.

Most resistors are mounted vertically. Use a uniform method of bending the long terminal twice to obtain right angles. Semicircles indicate where the resistors sit on the board. Where rectangles are printed, the resistor is mounted flat on the board. Although it's possible to solder all opamps directly on the board, you do so at a risk (see part 1), hence the prototype was built using turned-pin 8-way DIL sockets for all opamps. It is essential to use premium quality sockets here — don't be tempted to use cheap ones with dodgy spring contacts; it is false economy.

The tallest components on the board are capacitors C2, coil L1, relay RE1 and the screw terminal blocks for the loudspeaker and supply connections. The coil consists of 10 turns of 1 mm diameter (AWG18) enamelled copper wire (ECW) with an internal diameter of 20 mm. The coil windings should be spread evenly to obtain an overall length to suit the footprint on the board. Fit the coil first, then R106 at a height of 5-10 mm above the board surface and not touching L1 anywhere.

Although printed on the silk screen PCB overlay, capacitors C24, C25, C26 and C27 are not fitted. Instead, a single 1,000 µF, 63 V electrolytic capacitor is used, its terminals being inserted directly into terminal blocks K16/K17 (observe the polarity and provide the leads with sleeving for insulation). This should be a high quality electrolytic capacitor with low ESR. The modification proved necessary to get rid of unex-

pected distortion levels occurring at about 20 kHz in the initial design and was found to totally cure the problem. The amplifier specifications printed last month apply to the capacitor-modified version only. The four corners of the amplifier board have holes for PCB standoffs — we used 10 mm tall ones.

The finished amplifier board should be given a thorough visual inspection before taking into use. Did you get all the polarised components' orientations right? Are all solder joints beyond reproach? It often helps to have a friend look at the board. Figure 1 once again shows the amp board for your reference. See how close you can get to this level of sophistication in building high-end audio circuitry.

Construction — PSU board

This is also a classic board with nothing but standard components, hence should be easy to assemble and get working. It's just the heatsinking of the voltage regulators that requires some mechanical work.

The two bridge rectifiers B1 and B2 should be fitted with 2 mm thick 70 × 35 mm aluminium plates bolted directly to the flat sides. The two voltage regulators IC1 and IC2 are secured to a single black, finned, extruded aluminium heatsink of which the mounting outline is shown on the component overlay. The heatsink is secured to the board with three M4 screws or bolts, for which holes need to be drilled and tapped in the underside. Venting holes are provided in the PCB area under the heatsink

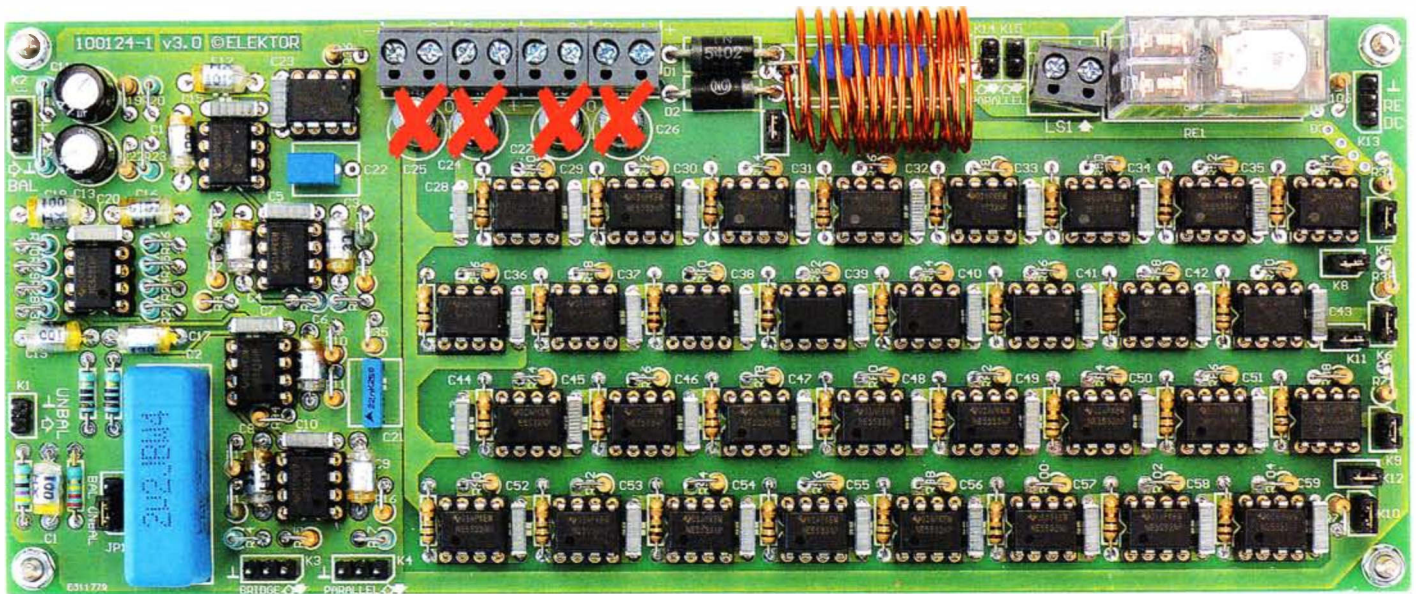


Figure 1. Perfect sound from perfect construction. Can you match it? Please note that capacitor positions C24–C27 must remain empty.

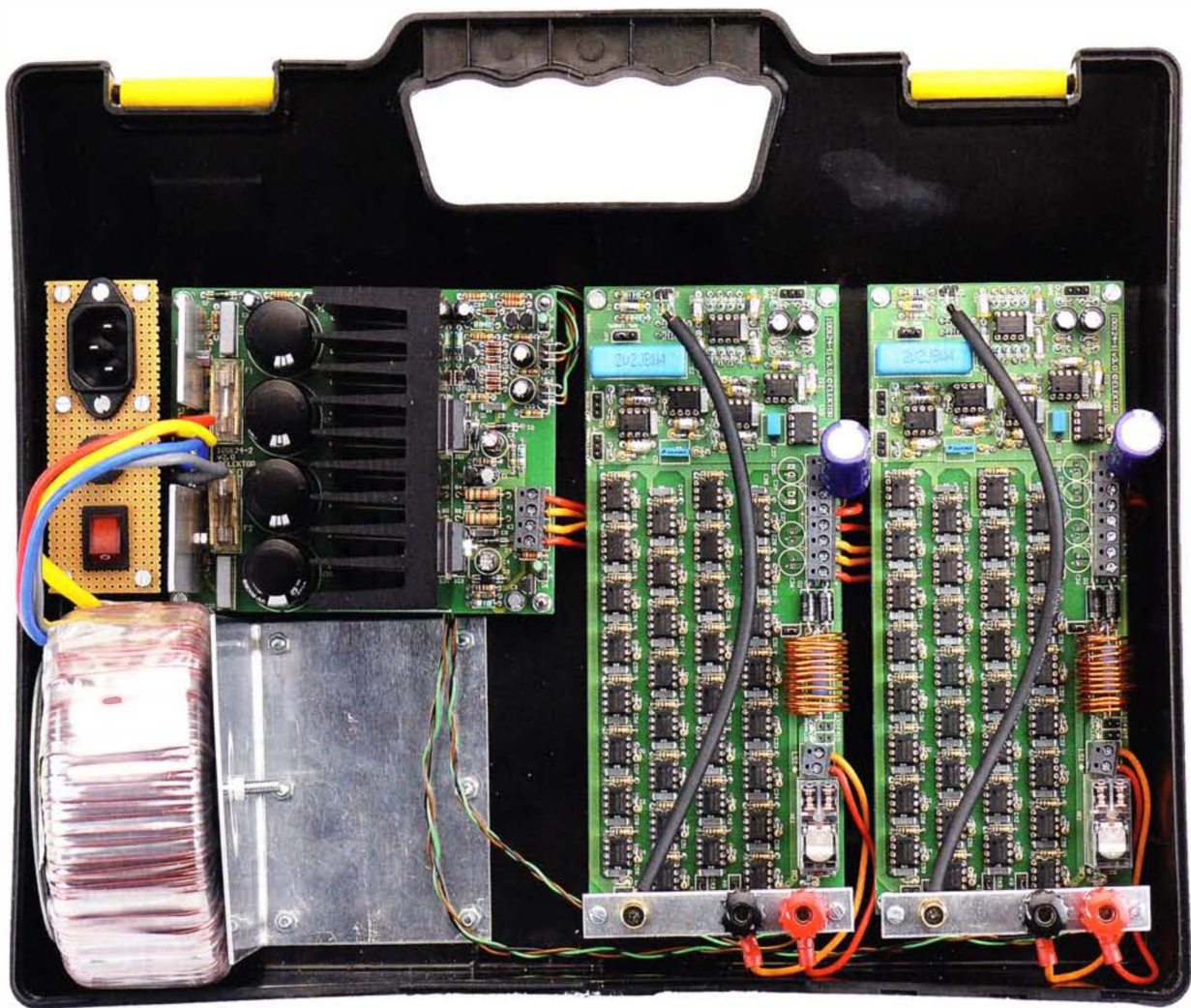


Figure 2. The power supply board is conventional as far as construction and assembly is concerned, but do make sure you get the mounting of the heatsink and the voltage regulators right.
An ABS suitcase as shown here is not recommended as a permanent housing for the amplifier.

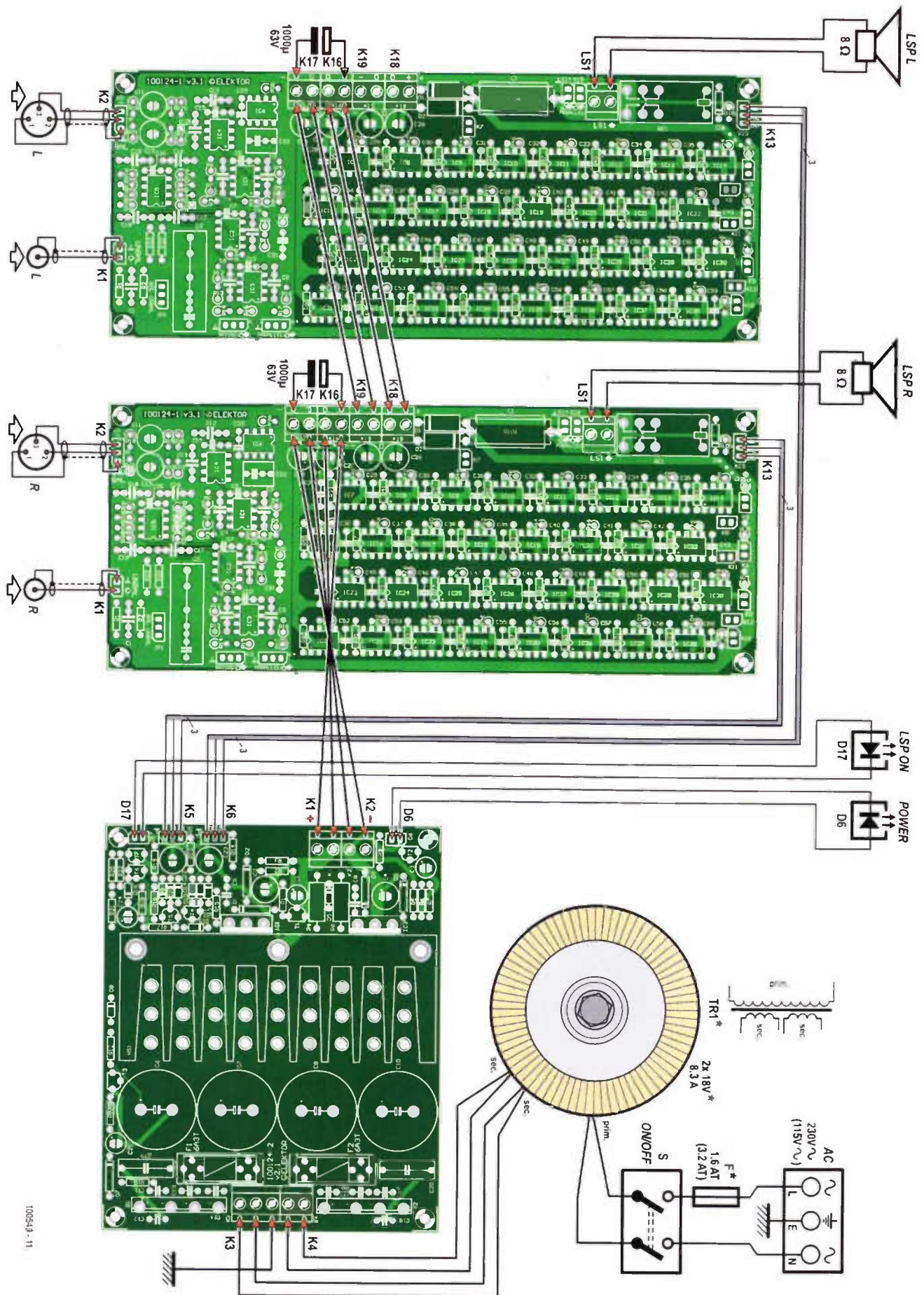


Figure 3. OpAmp interwiring diagram. Note that this is for the 2 x 15 watts, 8-ohm version. The electrical ratings of the toroidal transformer and the fuse should match your local AC line voltage. High amps and loudspeaker connections are shown in slightly thicker lines.

Masterclass #1: Extension of design for 4 ohm speakers

It is an inherent property of the 5532 power amplifier that its output current is limited by the internal overload protection of the opamps used. This means that if more current is needed in the load, you need to put more opamps in parallel. The basic design is intended to drive 8 Ohm loudspeakers with a reasonable safety margin, but as it stands it is definitely not recommended for 4 ohm operation.

To extend the 5532 amplifier for 4 ohm operation, it is necessary to double the number of 5532 sections driving the output. This is easier than it might sound because facilities are provided for adding one or more power amplifier PCBs in parallel. The connector K4 (see circuit schematic published last month) is used as an output from the main power amplifier PCB; it comes from the 'zero-impedance' stage IC3A, so the output is at a very low impedance (I measured 0.24 ohms at 1 kHz) but is immune to HF instability caused by cable capacitance. Any desired length of cable can be used. The output from K4 drives an identical power amplifier PCB that has the all the output opamps in place, but the redundant input amplifier circuitry

omitted. The equivalent connector K4 on this PCB is used as an input, and drives the output opamps in exactly the same way as on the main power amplifier PCB.

The connectors K14, K15 are used to connect together the outputs of the two amplifier PCBs. Note that this connection is made upstream of the output muting relay RE1, to preserve full protection for the loudspeaker in case of a DC fault.

While this conversion for 4 ohm operation is relatively straightforward, it is essential to remember that the power supply requirements are doubled along with the current output. You will need to consider a larger power transformer, more capable rectifiers, larger reservoir capacitors, and increased supply regulator capacity. In view of these various requirements, this 4 ohm version is mentioned here more as an option for the experimenter rather than a fully cut-and-dried design.

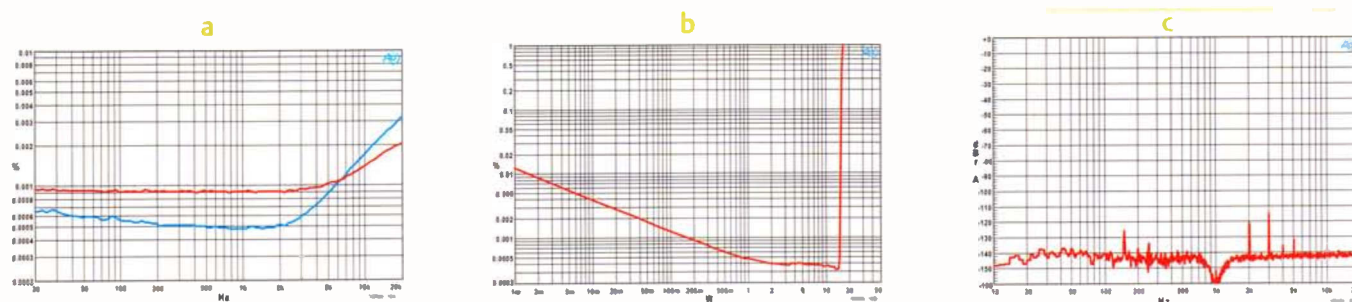


Figure 4. The OpAmpifier was duly 'grilled' on our AP System 2 analyser.
Graph (a): THD & noise vs. output power. Graph (b): distortion vs. output power (1 kHz, BW = 22 KHz).
Graph (c): FFT graph for 1 kHz, 1 watt, 8 ohms.

to assist cooling. The LT1083CP devices are secured to the heatsink with a heat conducting washer inserted. Two holes have to be drilled for the M3 bolts and nuts that hold the regulators firmly on the heatsink. First, determine the location of the holes for the regulators; then secure the heatsink to the board. The regulator terminals should be 'kinked outward' slightly for thermal relief. The two LT1083 regulators should be mounted, secured and soldered last.

Figure 2 shows the PSU board in a "travel & demo" version of the amplifier. PCB stand-

offs are used as with the amplifier board. The PSU can be tested (carefully) by temporarily connecting the secondaries of the toroidal transformer to the AC input terminals and checking for the correct output voltage of 18.0 VDC $\pm$ 0.7 V on each rail.

The complete amplifier

The OpAmpifier should be built into in a metal case observing all relevant precautions in respect of electrical safety, specifically with respect to earthing and the use of wiring carrying the AC grid voltage (230 V or 115 V). The size and shape of the

case will depend on the number of amplifier boards used, as well as the associated power supply, see the **insets on bridged operation and 4-ohm conversion**. Allow for quite some heat developing in the amplifier case, from the heatsink as well as from the NE5532s. All those milliwatts add up!

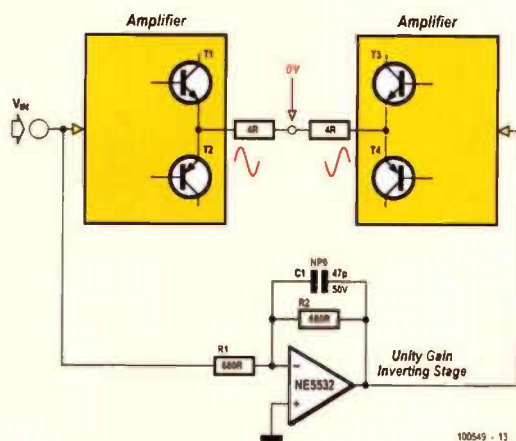
Everything ahead of the bridge rectifier AC terminals should be built, secured and wired with the high currents and voltages always in mind. If possible, do not extend the transformer wires. The amplifier inter-wiring diagram for the 8 ohm 2 x 15 watts

Masterclass #2: Extension of design for bridged operation

Two power amplifiers are bridged when they are driven with anti-phase signals and the load connected between their outputs; the load is not connected to ground in any way. This method of working doubles the signal voltage across the load, which in theory at least quadruples the output power. It is a convenient and inexpensive way to turn a stereo amplifier into a more powerful mono amplifier. Most conventional power amplifiers do not give anything close to power quadrupling – in reality the increase in available power will be considerably less, due to the power supply sagging and extra voltage losses in the two output stages, which are effectively in series. In most cases you will get something like three times the power rather than four times, and it may be less. I have come across many power amplifiers where the bridged mode only gave twice the output power, and it has to be said that in many cases the bridged mode looks like something of an afterthought.

The 5532 power amplifier will give better results than that for two reasons – firstly the power supply rails are regulated, and will not droop significantly under heavy loading. Secondly, the parallel structure minimises voltage losses; for example, all the 1 ohm output combining resistors are in parallel and their effective resistance is therefore very small.

Bridged mode is so called because if you draw the four output transistors of a conventional amplifier with the load connected between them, as in **Figure A**, it looks something like the four arms of a Wheatstone bridge.



In the drawing, the 8 ohm load has been divided into two 4 ohm halves, to emphasise that the voltage at their centre is zero, and so

as far as current output is concerned, both amplifiers are effectively driving 4 Ohms loads to ground. The current capability required is therefore doubled, with all that that implies for increased losses in the output stages. A unity-gain inverting stage is shown generating the anti-phase signal; there are other ways to do it but this is the most straightforward and it simply adds one more 5532 section to a design that already contains quite a few of them. The simple shunt-feedback unity-gain stage shown does the job very nicely, and the 5532 power amplifier incorporates a version of this circuit. The resistors in the inverting stage need to be kept as low in value as possible to reduce their Johnson noise contribution, but not of course made so low that the opamp distortion is increased by driving them. The capacitor C1 across the feedback resistor R2 assures HF stability – with the circuit values shown it gives a roll-off that is –3 dB at 5 MHz, so it does not in any way imbalance the audio frequency response of the two amplifiers.

You sometimes see the glib statement that bridging always reduces the distortion seen across the load because the push-pull action causes cancellation of the distortion products. It is not true. Push-pull systems can cancel even-order distortion products; odd-order harmonics will not be cancelled.

The 5532 power amplifier has facilities for bridging built in. The last stage in the input amplifier is the inverting stage around IC3A, which is needed to get the phase between amplifier input and output correct. If we take the signal off from before this stage, then it is inverted with respect to the amplifier output and can so be used to drive another power amplifier board in anti-phase; this PCB can be built with the input amplifier circuitry omitted, as for the 4 ohm conversion described above. This inverted signal is available at connector K3. The loudspeaker is connected between the two output terminals on the amplifier PCBs and the output ground terminals are not used.

As explained above, driving an 8 ohm load with two bridged power amplifiers means that each amplifier is effectively driving a 4 ohm load to ground, so to take full advantage of the bridging capability requires the output stages to be doubled up, so that there are two power amplifier PCBs in parallel driving in-phase, and two power amplifier PCBs in parallel driving in anti-phase. This is of course quite a serious undertaking, as the number of output opamps has been doubled for bridging, and then doubled again to give adequate output current capability. The power supply capability will also need to be suitably increased.

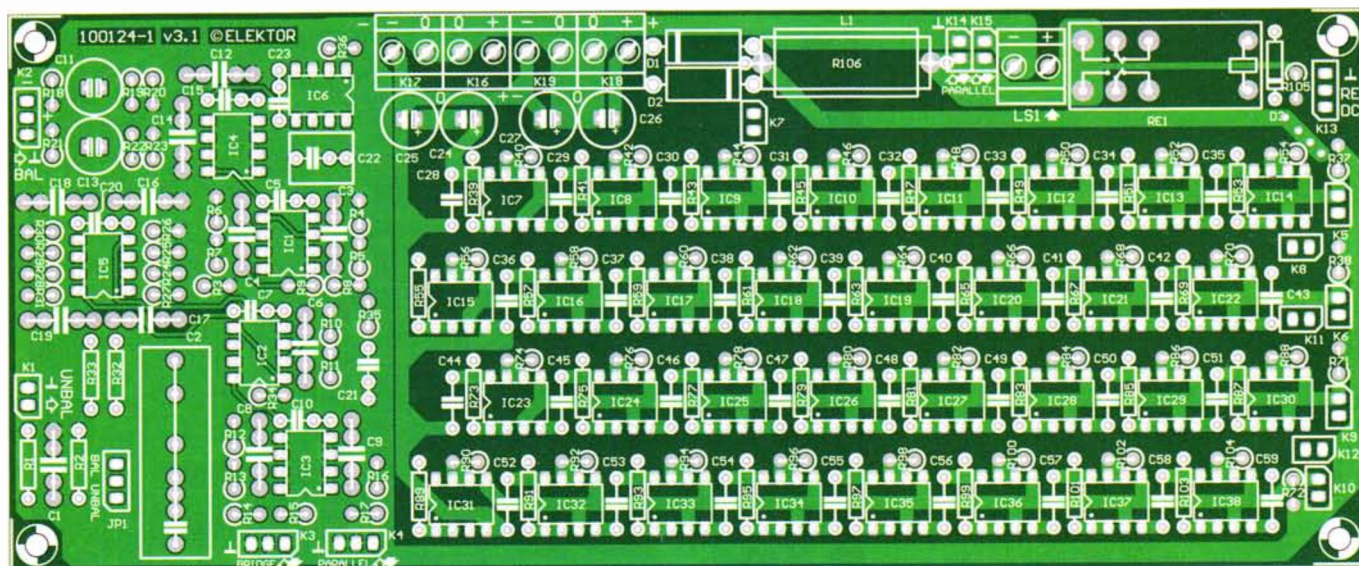
stereo version of the amplifier is shown in **Figure 3**. The rating of the AC power fuse should match the line voltage in your area (115 V/60 Hz or 230 V/50 Hz). Whichever, an approved IEC style appliance socket with integral fuseholder and double-pole rocker switch must be used.

We should emphasise that the 8.3-amp toroidal transformer indicated in the

drawing is advised for the 2 x 15 watt, 8-ohm version of the OpAmplifier. A much beefier transformer is required if you decide to build a 4-ohm or bridged version of the amplifier. As with nearly all audio power amplifiers, it's essential to have a good deal of spare capacity in the 'amps department' so do not skimp on the transformer.

The test results

The basic concept behind the 5532 power amplifier is to create an amplifier which has the very low distortion of the 5532 opamp. Preserving this performance when large currents are flowing to and fro on a circuit board is something of a challenge and requires very careful attention to grounding topology, supply-rail routing, and



COMPONENT LIST

Amplifier board (one channel)

Resistors

5%, 0.25W, Farnell/Multicomp MCF series, unless otherwise indicated.

R1, R14, R17 = 47Ω
 R2 = 220kΩ
 R3 = 47kΩ
 R4, R6 = 910Ω 1% 0.25W, Multicomp type MF25 910R
 R5, R7, R10, R11, R15, R16 = 2.2kΩ
 R8, R9, R32, R33 = 10Ω
 R12 = 1kΩ
 R13 = 2.00kΩ 1% 0.25W, Multicomp type MF25 2K
 R18, R21 = 150kΩ
 R19, R22 = 100Ω
 R20, R23 = 68kΩ
 R24-R31 = 820Ω
 R34, R35 = 10MΩ
 R36 = 100kΩ
 R37, R38, R71, R72 = 22kΩ

R39-R70, R73-R104 = 1Ω

R105 = 10kΩ

R106 = 10Ω 5% 3W, Tyco Electronics type ROX35J10R

Capacitors

C1, C12, C14, C16-C19 = 100pF, 2.5%, 160V, polystyrene, axial (LCR Components)
 C2 = 2.2μF 5% 250V, polypropylene, Evox Rifa type PHE426HF7220JR06L2
 C3, C4, C6, C8, C9 = 33pF ±1pF 160V, axial, polystyrene (LCR Components)
 C5, C7, C10, C15, C20, C23, C28-C59 = 100nF 10% 100V, lead pitch 7.5mm, Epcos type B32560J1104K
 C11, C13 = 47μF 20% 35V non polarised, Multicomp type NP35V476M8X11.5 (Farnell # 1236671)
 C21 = 22nF 10%, 400V, lead pitch 7.5mm, Epcos type B32560J6223K (Farnell # 9752366)
 C22 = 1μF, 10%, 10V, lead pitch 7.5mm, Epcos type B32560J1105K (Farnell # 9752382)
 C24-C27 = not fitted, see text

Inductors

L1 = 1.7μH; 10 turns 1 mm (18AWG) ECW,

diameter 20mm, effective length 20mm

Semiconductors

D1, D2 = 1N5402

D3 = 1N4148

IC1-IC5, IC7-IC38 = NE5532 (see text for notes on device selection)

IC6 = OP177 (Analog Devices)

Miscellaneous

K1, K5-K12, K14, K15 = 2-pin SIL pinheader, lead pitch 0.1 inch

K2, K3, K4, K13 = 3-pin SIL pinheader, lead pitch 0.1 inch

K16-K19, LS1 = 2-way PCB terminal block, lead pitch 5mm

JP1 = 3-pin SIL pinheader and jumper, lead pitch 0.1 inch

RE1 = PCB mount relay, PCB, DPCO, 24VDC/1100Ω/5A, Omron type G2R-2 24DC Phono socket, chassis mount, black, Neutrik type NYS367-0

XLR socket, chassis mount, 3-way, Pro Signal type PSG01588

PCB # 100124-1, Elektor Shop, see www.elektor.com/100549

decoupling arrangements. For this reason it is strongly recommended that you use the Elektor PCB design — better still, ready-made boards supplied by Elektor.

The completed prototype was measured using Elektor Labs' Audio Precision System Two Cascade Plus 2722 Dual Domain test-set, and Figures 4a, 4b and 4c show

the pleasing and impressive results. Figure 4a shows the harmonic distortion and noise as a function of signal frequency. Measurements carried out at 1 watt (red) and 8 watts (blue) of output power within 80 kHz bandwidth. The graph in Figure 4b shows distortion as a function of output power at 1 kHz and a bandwidth of 22 kHz. From about 3 watts the amplifier actu-

ally gets close to the lower measurement threshold and noise floor of our analyser! At 15 watts (roughly) the amplifier veers off into clipping. The curve in Figure 4c, finally, is an FFT plot for 1 kHz, 1 watt into 8 ohms. The fundamental frequency is suppressed. The second harmonic is way down at -121 dB and the third harmonic slightly higher at -115 dB. At a bandwidth

COMPONENT LIST

Power supply board

Resistors

5%, 0.25W, Farnell/Multicomp MCF series, unless otherwise indicated.

R1, R5 = 100Ω

R2, R6 = 1.30kΩ 1% 0.25W, Multicomp MF25 1K3

R3, R7 = 39Ω

R4, R8 = 1.5kΩ 5% 1W, Multicomp MCF 1W 1K5

R9, R11, R16, R20-R22 = 2.2kΩ

R10, R12-R15, R19 = 47kΩ

R17 = 150kΩ

R18 = 470kΩ

Capacitors

C1, C3, C24 = 47μF 20% 25V, lead pitch 2.5mm, Rubycon type 25ZLG47M6.3X7

C2, C4 = 100μF 20% 25V, lead pitch 2.5mm, 1.43A AC, Nichicon type UPM1E101MED

C5, C8 = 100nF 10% 100V, lead pitch 7.5mm, Epcos type B32560J1104K

C6, C7, C9, C10 = 4700μF 20% 35V, lead pitch 10mm, snap in, Panasonic type ECOS1VP-472BA (25mm max. diameter)

C11-C18 = 47nF 10% 50V ceramic, lead pitch 5mm

C19, C20 = 47nF 20%, 630VDC X2, lead pitch 15mm, Vishay BCcomponents BFC233620473

C21 = 4.7μF 20% 63V, lead pitch 2.5mm

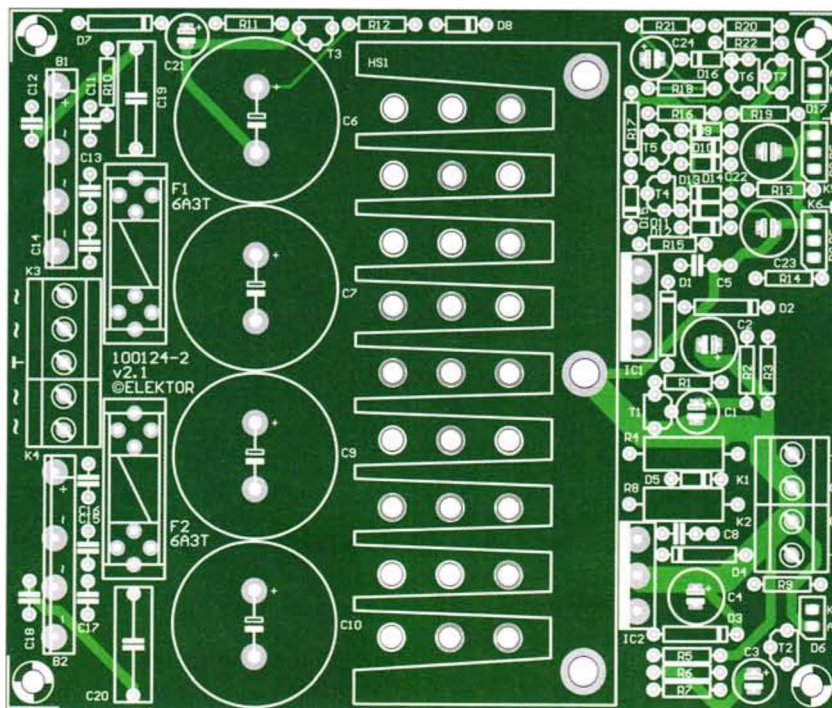
C22, C23 = 22μF 20% 35V, non polarised, lead pitch 2.5mm, Multicomp NP35V226M6.3X11

Semiconductors

D1-D4, D7 = 1N4002

D5, D8-D16 = 1N4148

D6, D17 = LED, green, 3mm



T1, T3, T4, T6 = BC337

T2, T5, T7 = BC327

IC1, IC2 = LT1083 (Linear Technology)

B1, B2 = GSIB1520 (15A/200V bridge rectifier) (Vishay General Semiconductor)

Miscellaneous

K1, K2, K4 = 2-way PCB terminal block, lead pitch 5mm

K3 = 3-way PCB terminal block, lead pitch 5mm

D6, D17, S1 = 2-pin SIL pinheader, lead pitch 0.1 inch

K5, K6 = 3-pin SIL pinheader, lead pitch 0.1 inch

F1, F2 = fuse, 6.3A antisurge (time lag), with 20x5mm PCB mount fuseholder and cover
Heatsink, Fischer Elektronik type SK92/75SA 1.6 K/W, size: 100x40mm, Farnell # 4621578, Reichelt # V7331G

Miscellaneous, mechanical

M3 screws, nuts and washers for mounting IC1 and IC2 to heatsink (see text)

TO-3P thermal pad, (Bergquist type K6-104)

M4x10 screws for mounting heatsink to PCB
PCB # 100124-2, Elektor Shop, see www.elektor.com/100549

of 22 kHz, the total level of noise and harmonics is about 0.0005%. If the distortion is measured for the two strongest harmonics only, it is at a level of 0.0002%.

Conclusion

To the non-initiated, the OpAmpifier with its case closed may be just another high-end audio amplifier albeit with quite impressive specs relative to the investment. To the discerning electronic engineer, it's a delightful and off-the beaten path approach to getting high-quality audio watts from a dead common component like the NE5532 you

normally associate with nano-amperes and microvolts electronics-wise, and pennies in the financial department!

The proof of the pudding is in the eating. That's why a demo version of the OpAmpifier was cased up in a rugged suitcase for easy transporting, packaging and demoing. At the time of writing, it's about to leave Elektor House on a journey along several audiophiles and audio communities around the world for listening tests. Their feedback is awaited eagerly.

(100549)

Spoilt for choice — which 5532?

It is an unsettling fact that not all 5532 opamps are created equal. The design is made by a number of manufacturers, and there are definite performance differences. While the noise characteristics appear to show little variation in my experience, the distortion performance does appear to vary significantly from one manufacturer to another. Although, to the best of my knowledge, all versions of the 5532 have the same internal circuitry, they are not necessarily made from the same masks, and even if they were, there would inevitably be process variations between manufacturers.

Since the distortion performance of the amplifier is unusually, and possibly uniquely, dependent on only one type of semiconductor, it makes sense to use the best parts you can get. In the course of the development of this project 5532s from several sources were tested. I took as wide a range of samples as I could, ranging from brand-new devices to parts over twenty years old, and it was reassuring to find that without exception, every part tested gave the good linearity we expect from a 5532. The information here will be of use not only in the 5532 power amplifier project, but should prove very valuable for anyone using 5532s in high-quality applications.

The main sources at present are Texas Instruments, Fairchild Semiconductor, ON Semiconductor, (was Motorola) NJR, (New Japan Radio) and JRC (Japan Radio Company). TI, ON Semi and Fairchild samples were compared in the Elektor labs by Ton Giesberts. The author for his part did THD tests on six samples from Fairchild, JRC, and Texas, plus one old Signetics 5532 for historical interest. The Elektor lab tests were carried out on an actual and very crucial section of the OpAmp design: the driver section! See **Figure A** for the circuit and **Figure B** for the AP2 plots.

As it turned out, the Texas 5532s (green line) proved to be distinctly inferior, both in the Elektor Labs and in the author's tests. We have to admit this surprised us, as we have always thought that the Texas part was one of the best available, but the measurements say otherwise. Distortion at 20 kHz ranges from 0.001% to 0.002%, showing more variation than Fairchild and ON Semi as well as being higher in general level. The low-frequency section of the plot, below 10 kHz, is approaching the measurement floor, as for all the other devices, and distortion is only just visible in the noise.

Compared with other maker's parts, the THD above 20 kHz is much higher—and at least 3 times greater at 30 kHz. Fortunately this should have no effect unless you have very high levels of ultrasonic signals that could cause intermodulation. If you have, then you have bigger problems on your hands than picking the best opamp manufacturer...

The graphs show undisputedly that the Fairchild 5532s (blue line) are top of the bill and true audio purists should select these devices even if they come at a slightly higher price as well as being awkward in terms of type code print on the device. Check with your supplier, it should be worth your while. There is effectively no distortion visible above the noise floor up to about 12 kHz, and distortion at 20 kHz is less than 0.0005%.

